

Splitting Cells in a Biosafety Cabinet

A standard procedure for passaging adherent cells, involving washing with PBS, detachment with Trypsin-EDTA, and reseeding into new culture vessels.

Equipment

- Biosafety cabinet (Sterile workspace for cell culture.)
- CO2 Incubator (Used for maintaining cell cultures at 37°C.)
- Vacuum aspirator (Used to remove waste media and PBS.)
- Pipette controller (Electronic device for serological pipetting.)
- 10cm culture dish (Vessel for growing adherent cells.)
- 15ml conical tube (Used for cell suspension transfer (protocol specifies 50ml).)

Reagents

- 70% ethanol (Used for sterilization of hands and materials.)
- PBS (Phosphate Buffered Saline used for washing cells.)
- 0.25% Trypsin-EDTA (Used to detach adherent cells from the dish surface.)
- Complete medium (Cell culture medium used to neutralize trypsin and provide nutrients.)

Procedure

1. Spray hands with 70% ethanol
2. Get all reagents
3. Spray all materials with 70% ethanol before placing in hood
4. Remove 10cm dish from incubator
5. Place dish in biosafety cabinet
6. Aspirate old medium completely using vacuum aspirator
7. Tilt dish to remove residual medium from edges
8. Add 3ml PBS to dish
9. Gently rock dish to wash cell surface
10. Aspirate PBS completely
11. Add 1ml of 0.25% Trypsin-EDTA to 10cm dish
12. Ensure even distribution by tilting dish
13. Gently tap sides of dish to help detachment
14. Add 5ml complete medium to dish
15. Pipette up and down 5-10 times to create single cell suspension
16. Transfer entire 6ml cell suspension to sterile 50ml conical tube
17. Calculate volume needed for 1:3 split (approximately 2ml of cell suspension)
18. Add 5ml cell suspension to new 10cm dish
19. Add 5ml fresh medium (total 10ml in dish)
20. Gently rock dish in cross pattern (North-South, East-West) to distribute cells
21. Place dishes back
22. Ensure dishes are level, clean-up, close hood